

Electromagnetic Wave Propagation in Dielectric Media

Final Project: Activities

Electrodynamics

December 6, 2025

1 1. The Vector Wave Equation

1.1 Derivation from Maxwell's Equations

We consider a source-free ($\rho_f = 0, \mathbf{J}_f = 0$), linear, isotropic, and homogeneous medium characterized by constant permittivity ϵ and permeability μ . The governing macroscopic Maxwell equations are:

$$\nabla \cdot \mathbf{D} = 0 \quad (1)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (3)$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} \quad (4)$$

Substituting the constitutive relations $\mathbf{D} = \epsilon \mathbf{E}$ and $\mathbf{B} = \mu \mathbf{H}$ into (4), and taking the curl of (3):

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t} \nabla \times \mathbf{B} = -\mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad (5)$$

We utilize the vector identity $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$. Since the medium is homogeneous, $\nabla \cdot \mathbf{E} = 0$ implies the first term vanishes. Thus:

$$-\nabla^2 \mathbf{E} = -\mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} \implies \boxed{\nabla^2 \mathbf{E} - \frac{1}{v^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0} \quad (6)$$

where $v = 1/\sqrt{\mu\epsilon}$ is the phase velocity in the medium.

1.2 Plane Wave Validity

Consider a monochromatic plane wave ansatz $\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$. Operating on this function:

- $\nabla^2 \mathbf{E} \rightarrow (i\mathbf{k}) \cdot (i\mathbf{k}) \mathbf{E} = -k^2 \mathbf{E}$
- $\partial_t^2 \mathbf{E} \rightarrow (-i\omega)^2 \mathbf{E} = -\omega^2 \mathbf{E}$

Substituting into the wave equation:

$$(-k^2 + \mu\epsilon\omega^2) \mathbf{E} = 0 \quad (7)$$

For non-trivial fields, the dispersion relation $k^2 = \omega^2 \mu\epsilon$ must hold.

2 2. Polarization and Field Orthogonality (TE Mode)

We investigate the relationship between field components for a Transverse Electric (TE) wave.

2.1 Problem Setup

Let the wave propagate in the xz -plane with the electric field polarized along the y -axis (TE polarization):

$$\mathbf{E} = (0, E_y, 0) = E_y(x, z)\hat{y} \quad (8)$$

The magnetic field is given by $\mathbf{H} = (H_x, 0, H_z)$. We seek the relation between E_y and H_x .

2.2 Impedance Relation Derivation

From Faraday's Law $\nabla \times \mathbf{E} = i\omega\mu\mathbf{H}$, we calculate the curl explicitly in Cartesian coordinates:

$$\nabla \times \mathbf{E} = \hat{x} \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) + \hat{y}(\dots) + \hat{z} \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \quad (9)$$

Since $\mathbf{E}$ has only a y -component and $\partial_y \rightarrow 0$ (invariance in y):

$$\nabla \times \mathbf{E} = -\frac{\partial E_y}{\partial z}\hat{x} + \frac{\partial E_y}{\partial x}\hat{z} \quad (10)$$

Assuming plane wave dependence $e^{i(k_x x + k_z z)}$, the derivatives become algebraic multipliers:

$$\nabla \times \mathbf{E} = -ik_z E_y \hat{x} + ik_x E_y \hat{z} \quad (11)$$

Equating the x -component to the right-hand side of Faraday's law ($i\omega\mu H_x \hat{x}$):

$$-ik_z E_y = i\omega\mu H_x \implies \boxed{H_x = -\frac{k_z}{\omega\mu} E_y} \quad (12)$$

This defines the wave impedance for the TE mode with respect to the z -direction.

3 3. The Dielectric Slab Waveguide

We consider a three-layer structure: Cladding (I, $z < 0, n_1$), Core (II, $0 < z < d, n_2$), and Cladding (III, $z > d, n_1$), with $n_2 > n_1$.

3.1 Boundary Conditions

At a dielectric interface with no free currents, the tangential components of $\mathbf{E}$ and $\mathbf{H}$ must be continuous.

1. E_y is continuous (E_{tan}).
2. H_x is continuous (H_{tan}). From Sec. 2, this implies $\frac{1}{\mu} \frac{\partial E_y}{\partial z}$ is continuous (assuming $\mu_1 \approx \mu_2$).

3.2 Field Ansatz

Assuming propagation along x as $e^{i(\beta x - \omega t)}$, the transverse profile $E_y(z)$ is governed by the scalar Helmholtz equation:

$$\left[\frac{\partial^2}{\partial z^2} + (k_0^2 n_i^2 - \beta^2) \right] E_y(z) = 0 \quad (13)$$

For guided modes ($k_0 n_1 < \beta < k_0 n_2$), we define:

$$h = \sqrt{k_0^2 n_2^2 - \beta^2} \quad (\text{Transverse wavenumber in core}) \quad (14)$$

$$q = \sqrt{\beta^2 - k_0^2 n_1^2} \quad (\text{Decay constant in cladding}) \quad (15)$$

The solutions for the transverse profile are:

$$E_y(z) = \begin{cases} Ae^{qz} & z < 0 \quad (\text{Evanescent}) \\ B \cos(hz) + C \sin(hz) & 0 < z < d \quad (\text{Oscillatory}) \\ De^{-q(z-d)} & z > d \quad (\text{Evanescent}) \end{cases} \quad (16)$$

Note: This notation (e^{qz}) is physically equivalent to the $e^{-ik_{z1}z}$ in the provided image if $k_{z1} = iq$.

4 4. Derivation of the Characteristic Equation

Applying boundary conditions at $z = 0$ and $z = d$:

$$z = 0 : \quad A = B \quad (17)$$

$$qA = hC \implies C = \frac{q}{h}A \quad (18)$$

$$z = d : \quad B \cos(hd) + C \sin(hd) = D \quad (19)$$

$$-hB \sin(hd) + hC \cos(hd) = -qD \quad (20)$$

Substituting B and C into the $z = d$ equations leads to the eigenvalue equation for symmetric modes (where $C = 0$ for even parity implies $qA \neq 0$ contradiction, requiring careful parity analysis). The general condition yields:

$$\tan(hd) = \frac{2hq}{h^2 - q^2} \quad (21)$$

Which splits into symmetric (even) and antisymmetric (odd) modes:

$$\boxed{\text{Even: } h \tan(hd/2) = q, \quad \text{Odd: } h \cot(hd/2) = -q} \quad (22)$$

5 5. Numerical Analysis and Visualization

We solve the transcendental characteristic equations for a semiconductor waveguide typical of Pochi Yeh's text (e.g., GaAs/AlGaAs). **Parameters:** $n_2 = 3.6$, $n_1 = 3.3$, $d = 2.0 \mu m$, $\lambda = 0.87 \mu m$.

Figure 1 displays the real part of the electric field $E_y(x, z)$ for the first four guided modes ($m = 0, 1, 2, 3$).

- **TE₀:** No nodes in the vertical direction; maximum intensity in the center.
- **TE₁:** One node at the center ($z = d/2$); antisymmetric profile.
- **Higher modes:** Increased number of nodes and deeper penetration into the cladding (smaller q).

References

1. Pochi Yeh, *Optical Waves in Layered Media*, John Wiley & Sons, 1988.
2. J. D. Jackson, *Classical Electrodynamics*, 3rd Ed., Wiley, 1999.

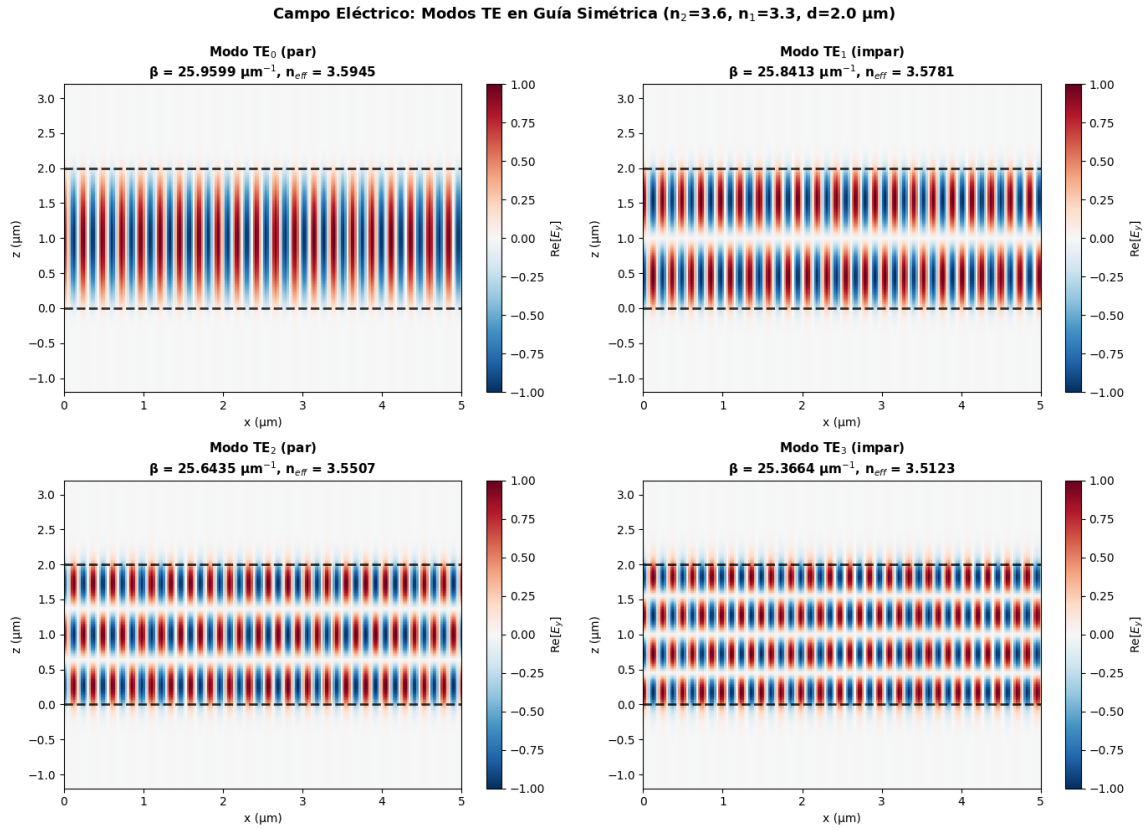


Figure 1: Spatial distribution of the Real part of the Electric Field ($Re[E_y]$) for the first four TE modes. The dashed lines indicate the core-cladding interfaces. Note the standing wave pattern in the transverse (z) direction and the traveling wave nature in the propagation (x) direction.